PHISHLENS

Phishing URL Detection System

Easy Hinglish Guide for Project Explanation

Ek line mein: PhishLens URL ke text se kuch useful numbers nikalta hai, phir machine-learning model se estimate karta hai ki URL phishing jaisa lagta hai ya nahi.

Project ko simple example se samjho

Jaise teacher kisi sentence ke words dekhkar uska meaning samajhte hain, waise hi model URL ke patterns dekhta hai. Jaise bahut lamba URL, ajeeb hostname, ya 'login' aur 'verify' jaise words. Model in patterns ko pehle se labeled examples se seekhta hai.

System ka flow

1. Dataset	2. URL features	3. ML model	4. App result
Pehle se labeled URLs	URL ko numbers mein badalna	Patterns se score banana	Risk score + reason dikhana

Important boundary

Yeh version URL ke text ko analyze karta hai. Yeh website open nahi karta, page ka content nahi padhta, aur kisi login form ko submit nahi karta. 'Not flagged' ka matlab guaranteed safe nahi hota.

Is guide mein kya milega

Dataset, features, dono algorithms ka idea, training aur testing ka process, result metrics, app, files, packages, teacher ke liye bolne ka script, aur viva ke sawal.

1. Dataset aur labels

Model examples se seekhta hai. Har row mein URL aur uska pehle se diya hua label hota hai.

Item	Project mein kya hai
Source	Kaggle Phishing Site URLs; bundled sample is a subset whose Hugging Face card credits Kaggle.
Rows	3,000 sample rows. Duplicate URLs hatane ke baad 2,998 unique URLs training mein.
Labels	1,500 legitimate (0), 1,498 phishing (1).
Sample ka fayda	File chhoti rehti hai aur laptop par training aasaan hoti hai.
Sample ki limit	Yeh balanced classroom sample hai. Asli internet par phishing aur legitimate sites ka ratio alag hota hai.

Label ka matlab: 0 = dataset mein legitimate; 1 = dataset mein phishing. Yeh label ka naam hai, site ki hamesha ki safety guarantee nahi.

2. URL features - pehla group

Feature ka simple matlab: URL ke baare mein nikala hua ek measurable signal. Model in signals ko ek saath dekhkar decision seekhta hai.

Feature	Aasaan matlab
url_length	Poore URL mein total characters.
hostname_length	Website ke host/domain naam ki length.
path_length	Domain ke baad webpage ka path kitna lamba hai.
query_length	Question mark ke baad URL parameters ki length.
count_dots	URL mein dots ki ginti.
count_hyphens	Hyphen (-) ki ginti; bahut saare hyphens ek signal ho sakte hain.
count_at	@ symbol ki ginti.
count_question	? symbol ki ginti.
count_ampersand	& symbol ki ginti; query parameters ko separate karta hai.
count_equal	= symbol ki ginti; parameter names aur values mein hota hai.
count_digits	URL mein total numeric characters.

3. URL features - doosra group

Yeh baaki signals hostname structure, suspicious wording aur character patterns ko count karte hain.

Feature	Aasaan matlab
count_subdomains	Main domain se pehle kitne extra naam hain; jaise account.example.com mein account subdomain hai.
has_ip_address	Hostname mein normal naam ke badle IP number hai ya nahi: 1 = haan, 0 = nahi.
has_port	URL mein custom network port number diya hai ya nahi.
has_suspicious_word	URL mein login, verify, secure, account jaise selected words hain ya nahi.
suspicious_word_count	Selected suspicious words total kitni baar aaye.
has_percent_encoding	URL mein percent sign (%) hai ya nahi; kuch characters encoded ho sakte hain.
has_double_slash_in_path	Path ke andar extra // hai ya nahi.
tld_length	Top-level domain suffix ki length; jaise .com ka suffix 3 letters ka hai.
letter_ratio	Total URL characters mein letters ka fraction.
digit_ratio	Total URL characters mein digits ka fraction.
url_entropy	Characters kitne mixed ya unpredictable lagte hain; random-looking text ka signal.

Ek URL ka feature example

URL: https://secure-paypal-login.example.com/verify

Is URL mein 'secure', 'login' aur 'verify' jaise words hain, hostname lamba hai, aur hyphens hain. Feature extractor in baaton ko numbers mein badalta hai. Ek feature akela decision nahi karta; model sab features ka pattern dekhta hai.

Feature signals hain, proof nahi. Legitimate websites mein bhi long URLs ya 'login' word ho sakta hai.

Feature extraction code kahan hai?

url_features.py URL ko normalize karta hai, host/path/query parts parse karta hai aur ek table row jaisa numeric result deta hai. Training aur app dono isi function ko use karte hain, taaki feature ka meaning dono jagah same rahe.

4. Algorithms ka idea

Project do models compare karta hai: ek simple baseline aur ek tree ensemble.

Logistic Regression - baseline

Naam mein Regression hai, lekin yahan ise classification ke liye use karte hain. Yeh features ko weighted importance deta hai aur phishing class ka estimated score banata hai. Is project mein pehle StandardScaler features ko comparable scale par laata hai. Isse URL length jaise bade numbers model ke comparison ko bekaar tareeke se dominate nahi karte.

Random Forest - selected model

Random Forest bahut saare decision trees ka group hai. Har tree training examples se alag rules seekhta hai aur phishing ya legitimate ke liye vote deta hai. Forest votes se estimated score banata hai. Is run mein 300 trees hain; minimum leaf size 2 rakhi gayi. Class weighting bhi di gayi taaki class frequency ka asar manage ho.

Simple analogy: ek hi examiner par bharosa karne ke badle, kai decision trees vote karte hain. Random Forest un votes ko combine karta hai.

Data ko teen parts kyun banaya?

Part	Rows	Kis kaam ke liye?
Train	1,821	Model yahan se patterns seekhta hai.
Validation	555	Models compare aur alert threshold choose karne ke liye.
Test	622	Aakhri, alag evaluation ke liye; model select karne ke liye nahi.

Domain split

Same registrable domain ke URLs ko ek hi group mein rakha gaya. Isse ek domain ke milte-julte URLs train aur test dono taraf jaakar model ko website yaad karne ka shortcut nahi dete. GroupShuffleSplit is grouping ko random_state 42 ke saath reproducible tareeke se karta hai.

Threshold

Model score ko alert decision mein badalne ke liye threshold lagta hai. Is run ka threshold 0.406 hai: score 40.6% ya zyada ho toh phishing flag hota hai. Code validation data par precision-recall curve check karta hai aur F1 sabse achha karne wala threshold leta hai. Isliye threshold zaroori nahi 0.5 ho.

5. Result ko kaise samjhein

Final metrics sirf held-out domain test split se hain. Is result run mein Random Forest select hua.

Metric	Result	Aasaan matlab
Accuracy	79.1%	622 test URLs mein total kitne sahi nikle.
Precision	72.3%	Jin URLs ko phishing flag kiya, unmein kitne waqai phishing label wale the.
Recall	85.0%	Dataset ke phishing URLs mein se kitne model ne pakde.
F1	78.1%	Precision aur recall ka combined balance score.

Confusion matrix

Rows actual class hain, columns model ki prediction. Class order pehle 0 legitimate, phir 1 phishing hai.

Actual / predicted	Legitimate (0)	Phishing (1)
Legitimate (0)	260 sahi	89 galat alert
Phishing (1)	41 chhoot gaye	232 sahi pakde

Iska seedha matlab

Test set mein 273 phishing URLs the. Model ne 232 pakde aur 41 miss kiye. Saath hi 349 legitimate URLs the, jinmein 89 ko galti se phishing flag kiya. Isliye recall zyada ho sakta hai, jabki precision kam - dono alag sawal ka jawab dete hain.

Accuracy formula: (sahi legitimate + sahi phishing) / total = (260 + 232) / 622 = 79.1%.

Precision formula: sahi phishing alerts / sab phishing alerts = 232 / (232 + 89) = 72.3%.

Recall formula: pakde gaye phishing URLs / total phishing URLs = 232 / (232 + 41) = 85.0%.

F1 precision aur recall ka harmonic average hai. Yeh ek hi number mein dono ko balance karke dikhata hai.

Yeh balanced 3,000-row sample aur ek held-out split ka classroom result hai. Asli duniya mein URL patterns aur class ratio badal sakte hain.

Example app prediction

Included model ne reserved example URL https://secure-paypal-login.example.com/verify ko 94.7% phishing score diya; training threshold 40.6% hone ki wajah se use flag kiya. Yeh reserved example domain hai aur is result ka matlab real attack ya proof nahi.

6. App, SHAP aur project files

App Streamlit par chalti hai. User ek URL deta hai aur trained model result dikhata hai.

Step	App mein kya hota hai
1	app.py model.joblib load karti hai.
2	User Website URL field mein URL likhta hai.
3	Wahi url_features.py us URL ke features banata hai.
4	model.predict_proba phishing label ka estimated score nikalta hai.
5	Score threshold se compare hota hai; app flag ya not flagged result dikhati hai.
6	SHAP top feature contributions dikhata hai, bade contribution pehle.

SHAP ko simple tareeke se samjho

SHAP model ki prediction ko feature contributions mein todta hai. Is project mein ek chhota background sample training rows se liya gaya, phir permutation explainer feature values ko alag-alag karke dekhta hai. App phishing score ko kitna upar ya neeche move kiya, us hisaab se top 10 features dikhati hai. Yeh model explanation hai, cause ka proof nahi.

Project files

File	Role
data/processed/phishing_small.csv	Bundled, compact URL dataset.
prepare_data.py	Full Kaggle CSV mile toh clean karke stratified sample bana sakta hai.
url_features.py	URL ko numeric model features mein badalta hai.
train_model.py	Split, train, compare, threshold selection aur test metrics.
model.joblib	Selected trained model, threshold, feature names, SHAP background sample.
app.py	Streamlit URL review interface.
results.json	Metrics, split sizes, threshold aur confusion matrix.
requirements.txt	Project ke Python packages aur versions.

Main packages

pandas CSV/table data ke liye; scikit-learn models, split aur metrics ke liye; NumPy numeric calculations ke liye; joblib model save/load ke liye; Streamlit app ke liye; SHAP explanation ke liye; tldextract domain parts ke liye. Python ke standard tools json, pathlib, re, math, ipaddress aur urllib.parse report, file paths aur URL parsing mein madad karte hain.

7. Teacher ko samjhane ka script

Is script ko apne bolne ke style mein practice karna. Technical words ke baad apni simple explanation dena.

Mera project PhishLens hai. Yeh machine learning ka phishing URL detection system hai. Iska goal URL ke pattern dekhkar phishing ya legitimate ka estimated result dena aur prediction ke important reasons dikhana hai.

Maine Kaggle ke Phishing Site URLs dataset ka chhota 3,000-row sample use kiya. Duplicate URLs hatane par 2,998 unique URLs rahe. Label 0 legitimate aur label 1 phishing hai. Dataset mein page content nahi hai, isliye system URL text ke features hi use karta hai.

Pehle feature extraction hota hai: URL length, punctuation, hostname structure, suspicious words aur character ratios jaise numbers nikalte hain. Maine Logistic Regression ko simple baseline aur Random Forest ko doosra model rakha. Random Forest kai decision trees ke votes combine karta hai.

Data ko domain ke hisaab se train, validation aur test groups mein split kiya, taaki same website training aur test dono mein na aaye. Validation data par F1 se model aur alert threshold choose kiya. Final test split par accuracy 79.1%, precision 72.3%, recall 85.0% aur F1 78.1% mila.

Streamlit app URL ka score aur decision dikhati hai. SHAP batata hai kaun se URL features ne score ko influence kiya. System website ko open nahi karta. Yeh classroom prototype hai; not flagged ka matlab guaranteed safe nahi hota.

Viva ke aam sawal

Sawal	Simple jawab
Phishing kya hai?	Fake ya dhokhe wali website/URL jo sensitive details lene ki koshish kare.
Yeh website kholta hai?	Nahi. Sirf URL text ko parse karta hai; crawl ya login nahi karta.
Random Forest kyun?	Kai trees ke votes combine karta hai aur URL features ke mixed patterns seekh sakta hai.
Threshold 0.5 kyun nahi?	Validation set par F1 improve karne wala threshold 0.406 mila.
Recall aur precision mein farq?	Recall: kitne phishing URLs pakde. Precision: flagged URLs mein kitne waqai phishing the.
Domain split kyun?	Same website ke URLs train/test mein mix na hon aur memorization ka risk kam ho.
SHAP kya batata hai?	Har feature ne model score ko estimate ke mutabik kitna upar/neeche kiya.
Kya safe prediction guarantee hai?	Nahi. Model galti kar sakta hai; not flagged guaranteed safety nahi.
Current limitation?	Sample chhota aur balanced hai; sirf URL features hain, webpage content nahi.

Aakhri yaad rakhne wali baat

Teacher ko seedha batao ki yeh model phishing risk ka signal deta hai, final security proof nahi. PPT ke larger webpage-feature proposal se is version ka scope alag hai kyunki runnable project ke liye chhota Kaggle-derived URL dataset chuna gaya.